Multiscale sample entropy analysis of EEG data reveals spatially-localised differences between normal and Alzheimer's Disease and frontotemporal dementia brain activity

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1 Introduction

Alzheimer's Disease (AD) is a progressive neurodegenerative disorder and the most prevalent form of dementia. It is characterised by the accumulation of amyloid plaques, neurofibrillary tangles, neuropil threads, and dystrophic neurites, which lead to synaptic dysfunction and neuronal loss. [1] The cardinal manifestations of AD include memory impairment, cognitive decline, and behavioural abnormalities thus exerting a profound impact on the well-being of patients and severely compromising their quality of life.

The diagnosis of AD entails a comprehensive assessment involving multiple medical modalities. Neurophysiologists administer the Mini-Mental State Examination (MMSE), a cognitive questionnaire, to evaluate the mental status of individuals, employing a 30-point scale where lower scores indicate more pronounced cognitive deterioration. Additionally, neuroimaging techniques such as magnetic resonance imaging (MRI) or computed tomography (CT) and electroencephalography (EEG) recordings are employed alongside general neurological and physical examinations. [2]

As EEG represents a cost-effective diagnostic tool compared to the other methods, researchers have focused on analyzing EEG recordings derived from AD patients to uncover potential biomarkers related to the complexity, synchrony,

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and regularity of EEG activity. Experimental protocols involve acquiring EEG data from patients in different clinical settings of rest or cognitive tasks, and analysing such data with signal processing techniques aimed at extracting discriminative features, to differentiate between AD patients and healthy controls or draw comparisons with age-matched healthy subjects. [3] [4]

Among the various mathematical representations of EEG recordings that can be obtained via different signal processing techniques, complexity has been proved to be very effective in capturing sophisticated non-linear discriminative features between healthy subjects and AD patients. Previous studies have examined diverse measures of complexity, including auto-mutual information [5], correlation dimension [6], largest Lyapunov exponent [6], approximate entropy [5], permutation entropy [4], sample entropy [7, 8], Tsallis entropy [9], and Lempel-Ziv complexity [7]. Findings have consistently revealed disturbances in EEG complexity, with moderate AD patients displaying a decline in physiological EEG complexity compared to age-matched healthy individuals.
In this essay, I will use the multiscale sample entropy metric [10] to investigate whether there are clear statistical differences in the spatial distribution of complexity among the different EEG sources between normal control (NC), fronto-temporal dementia (FtD) and AD subjects.

2 Materials and methods

Multiscale sample entropy

In information theory, the Shannon's entropy $H(X)$ of a single discrete random variable X is a measure of its average uncertainty, and is computed as:

$$H(X) = -\sum_{x_i \epsilon \Theta} p(x_i) * \log p(x_i) = -E[\log p(x_i)],$$

with Θ being the set of values of the random variable X. As the entropy of dynamical, time varying systems (hence "real-world" data) is more complex to compute than stationary ergodic processes, different entropy metrics and their numerical approximations have been defined over the past decades. [11]
Among those, sample entropy stands as a fairly simple metric that shows little dependency over time series length and consistency over a broad range of parameters.
Let $X_i = x_1, ..., x_i, ..., x_n$ be a discrete time series of length N. From the time series, different m-length vectors can be extracted as $u_m(i) = x_i, x_{i+1}, ..., x_{i+m-1}$, $1 \leq i \leq N-m+1$. Now, let $n_i^m(r)$ be the number of vectors $u_m(j)$ that are close to the vector $u_m(i)$

(thus, $d[u_m(i), u_m(j)] \leq r$ with d being the Chebyshev distance): then, we can define the probability that of any vector $u_m(j)$ being close to vector $u_m(i)$ as $U_i^m(r) = \frac{n_i^m(r)}{N-m+1}$. The sample entropy is defined as:

$$S_E(m, r, N) = -ln\frac{U^{m+1}(r)}{U^m(r)}.$$

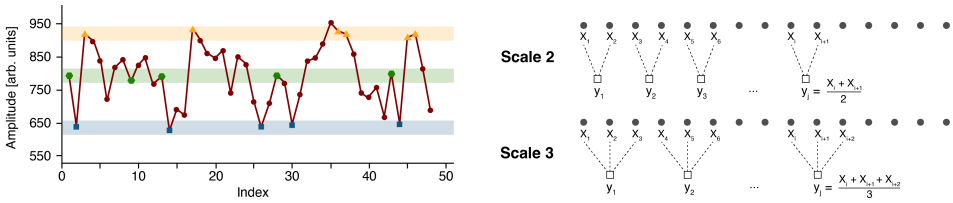
The sample entropy is therefore equal to the negative value of the natural logarithm of the conditional probability that sequences close to each other for m data points will also be close to each other when one more point is added to each sequence. A visual representation is shown in Figure 1a.
Costa et al. [10] introduced a refined procedure, called multiscale sample entropy (MSE), that, given a random one-dimensional time series X, computes the sample entropy values S_E for different coarse-grained realisation of the time series. The coarse-graining procedure is shown in Figure 1b: for each time scale factor τ, the coarse-grained process $y^{(\tau)}$ is computed as the average of non-overlapping windows of length τ of the initial time series X:

$$y_j^{(\tau)} = \frac{1}{\tau}\sum_{(j-1)\tau+1}^{j*\tau} x_i, 1 \leq j \leq \frac{N}{\tau}.$$

MSE is based on the observation that the output of complex systems is far from being regular and completely random: such systems generally reveal structures with long-range correlations on multiple spatial and temporal scales. MSE has been designed to address these multiscale features that are generally ignored by conventional entropy calculations.

2.1 Data

The dataset has been acquired from OpenNeuro [12], and it consists of EEG recordings from 88 subjects, 36 of which were diagnosed with Alzheimer's disease (AD group), 23 with Frontotemporal Dementia (FTD group) and 29 are healthy subjects (CN group). The datasets also reports the corresponding MMSE test outcomes for each subject. [13] The electrophysiological signals were recorded using the 19 scalp electrodes following the 10-20 international system, using a sampling rate of 500 Hz. The dataset contains both the original raw data and the pre-processed signals that have been analysed in the corresponding studies. However, since the authors performed ICA to filter the noisy EEG signals, in this essay I started from the raw data as ICA and other types of intensive processing steps can interfere with the underlying non-linear dynamics of the time series.

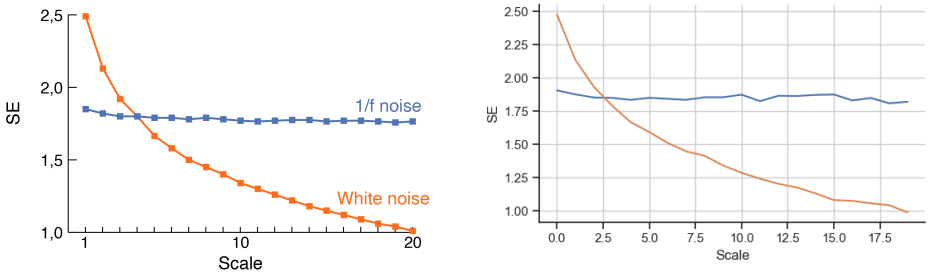


(a) Sample entropy computation shown on a simulated time series.

(b) Schematic illustration of the coarse-graining procedure

Figure 1: **Multiscale sample entropy (MSE).** The MSE is computed by taking the sample entropy (left, a) of different coarse-grained realisations at multiple scale factors τ (right, b) of the same temporal series X. On the left, dotted horizontal lines around data points represent tolerance values r. Two data points match each other, if the absolute difference between them is $\leq r$. Considering the two-component sequence $u[1], u[2]$ and the three-component template sequence $u[1], u[2], u[3]$, it can be seen that the two-component sequence is matched twice, while the three-component one is only matched once. These calculations are repeated for the next two-component and three-component template (namely, $u[2], u[3]$, $u[2], u[3], u[4]$). The number of sequences that match each of the two- and three-component template sequences are summed and added to the previous values. This procedure is then repeated for all other possible template sequences. SE is the natural logarithm of this ratio and reflects the probability that sequences that match each other for the first two data points will also match for the next point. Adapted from [10]

[14]
Following the procedures found in other EEG complexity studies [15], the raw data has only been filtered in the band of interest ($0,5-45Hz$, avoiding power line noise at $50Hz$), segmented in epochs and stripped of epochs containing eye-movement artefacts and other source of noises prior to computing the MSE.
As the MSE computation on such a huge dataset (each subject recording lasts more than 5 minutes) yields a prohibitive computational time, the entropy values were pre-computed and stored before the statistical analysis, using a custom Python script (see subsection 2.3).



(a) White and pink noise entropy from the original study.

(b) White and pink noise entropy replicated in this study.

Figure 2: **MSE computation comparison**. The figure shows the MSE values reported in the original study and the values obtained from the Python code used to compute the MSE in the current study. The orange line corresponds to the white noise MSE values, the blue one corresponds to pink noise MSE values plotted against the scale factor (from 1 to 20). The parameters used in both figures are $m = 2, r = 0.15, N = 30000$, with N corresponding to the sample points in both noises time series. Adapted from [10]

2.2 Statistical analysis

A statistical analysis was carried out to compare the distributions of the MSE values between the three groups (Alzheimer's (A), control (C) and frontotemporal dementia (F) patients). To do so, I first checked whether the requirements for multiple one-way ANOVA tests were satisfied (namely, the homoscedasticity of the data from the three groups) with a Bartlett's Test for each electrode site at each scale. As the ANOVA requirements couldn't be fulfilled (see section 3), the comparison between the group was carried out with multiple Kruskal-Wallis tests, computed once again for each electrode site at each scale. No interaction was investigated, but rather the tests were carried out to see if the group subdivision was statistically significant enough to distinguish the MSE values of the 88 subjects.

2.3 Software

The analysis was carried out using both Python and R Studio.
Python has been used to handle the original EEG data, filter it and compute the subjects MSE values. Loading the raw data and the filtering process have been implemented using the MNE library [16]. The MSE computation was adapted from

a MATLAB code and tested on dummy signals, such as white and pink noise, to verify the correctness of the algorithm. As it can be seen in Figure 2, the code was able to replicate the figure from the original MSE paper [10].

3 Results and discussion

Since ANOVA tests can only be significant if the variances of the groups are comparable (i.e, if the groups are homoscedastic), I first checked the variances of the groups using a Bartlett's test. As can be seen in Figure 3, some electrodes had heteroscedastic MSE values at different temporal scales.
Thus, I carried out the non-parametric equivalent of the one-way ANOVA test, the Kruskal-Wallis test (K-W test). A K-W test was computed for each electrode site at each temporal scale, comparing 380 sets of 88 values, to investigate which electrode-scale combination proved to be significant in discriminating the healthy patients from the pathological subjects. Indeed, not all of the electrodes sites have statistical differences in their MSE values among groups, but, very surprisingly, such differences would appear to be highly specific, being localised in the posterior-most electrode sites. Moreover, 4 out of 19 electrodes (O1-O2-T5-P3) preserve a p-value < 0.05 (i.e., reject the hypothesis of equal distributions) at every scale, and these electrodes are all located in the lower-left corner of the EEG topographic map (Figure 4).
The occipital electrodes O1-O2 are the electrodes with the lowest p-values across all scales: as Kruskal-Willis test only shows statistical differences among the means of three or more distributions but not which distributions differ, I further investigated the MSE distribution in these two electrodes. No significant difference is found between the two pathological groups, but both groups are statistically different from the control cases (Figure 5).
Overall these results suggest a spatially-localised difference in the brain activity of control cases compared to both AD and FtD patients during resting state. Specifically, it would appear that both pathologies decrease the signal complexity in the posterior-occipital areas. This is consistent with different studies that investigate the default mode network (DMN) disruption in AD. The DMN is a functional brain network whose areas show synchronous activity at resting-state functional magnetic resonance imaging when an individual is awake but not involved in any specific mental activity. Such areas include the medial prefrontal cortex (mPFC), posterior cingulate cortex (PCC), and precuneus [17]. Ingala et al. [18] showed that localized amyloid-β accumulation in posterior cingulate cortex (PCC) has a synergistic effect with DMN connectivity in predicting longitudinal memory

decline, which is consistent with the location of the brain activity differences highlighted in this essay.
Moreover, Grieder et al. [19] report a small reduction of functional connectivity between PCC and hippocampal areas in both hemispheres, and a constant reduction of DMN-MSE towards higher scales (which highlights a disturbance of signal complexity in both local and distal information processing), also shown in these results.
Finally, Puttaert et al. [20] characterised the power activation of the posterior DMN in AD patients, observing a significant reduction in the temporal stability of power activity within the posterior DMN, suggesting that it might represent an electrophysiological correlate of the decrease in mind wandering episodes previously reported in AD. The posterior DMN is indeed composed by the areas below some of the electrodes showing greater differences in Figure 4.

4 Conclusion

The present essay analysed a public dataset employing a different methodology compared to its original study. By extracting MSE values for each subject at each electrode site using 20 different temporal scales, the study observed statistically significant, highly-localised differences in signal complexity (i.e., information content) between healthy and pathological subjects during a resting-state experiment. Similar conclusions that have been shown in recent studies regarding AD and the DMN. This study proves the usefulness and the illustrative power of employing complexity measures (such as entropy) to test and verify hypotheses regarding intricated neural mechanisms from EEG data.

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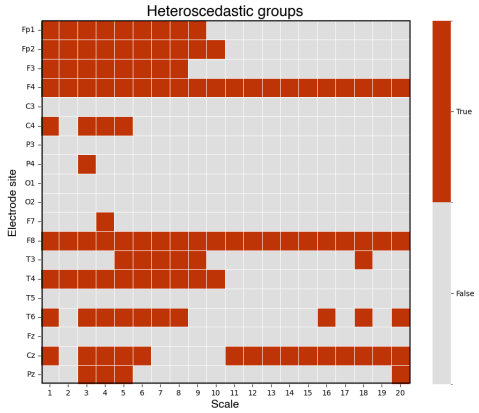
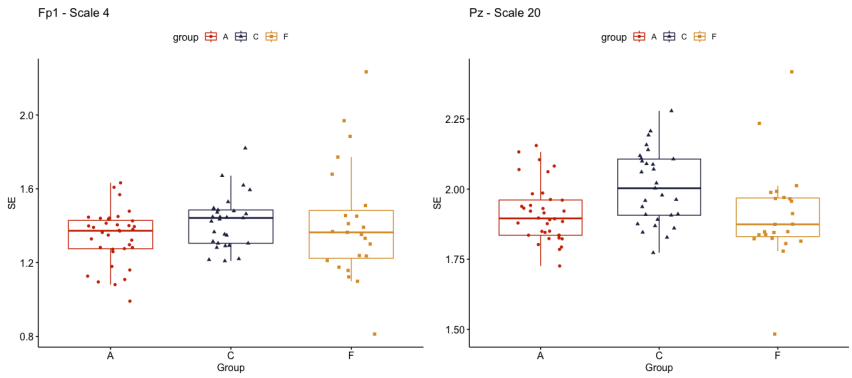
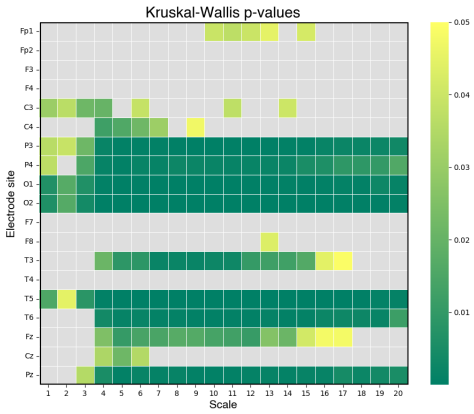


(a) Heteroscedasticity control over all temporal scales and electrode sites.

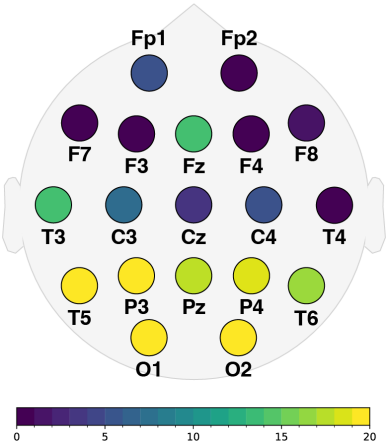


(b) SE values distribution for two heteroscedastic cases.

Figure 3: **Bartlett's test results.** The top figure shows the outcomes of each Bartlett's test carried out for each electrode site at each scale, where 'True' indicates a p-value < 0.05 (i.e., rejection of the null-hypothesis of the groups having equal variances). Each red square in the heathmap, therefore, represents an electrode-scale combination violating the ANOVA test homoscedasticity requirement. The bottom panel shows an example of how the values are distributed in two heteroscedastic cases: it can be seen that the height of the box plot greatly varies among the three groups (represented by the different colors).



(a) K-W test p-values over all temporal scales and electrode sites.



(b) Spatial representation of statistical differences among electrodes

Figure 4: **K-W test outcomes**. The top figure shows the p-values of each K-T test carried out for each electrode site at each scale. Gray squares indicate a p-value > 0.05 (i.e., rejection of the null-hypothesis of the groups having different means). The bottom panel summarise the matrix and plots its against the electrode locations: each location is coloured according to its number of statistically significant scales. The spatial distribution of brain activity differences show local effects of AD and FtD in the resting-state condition.

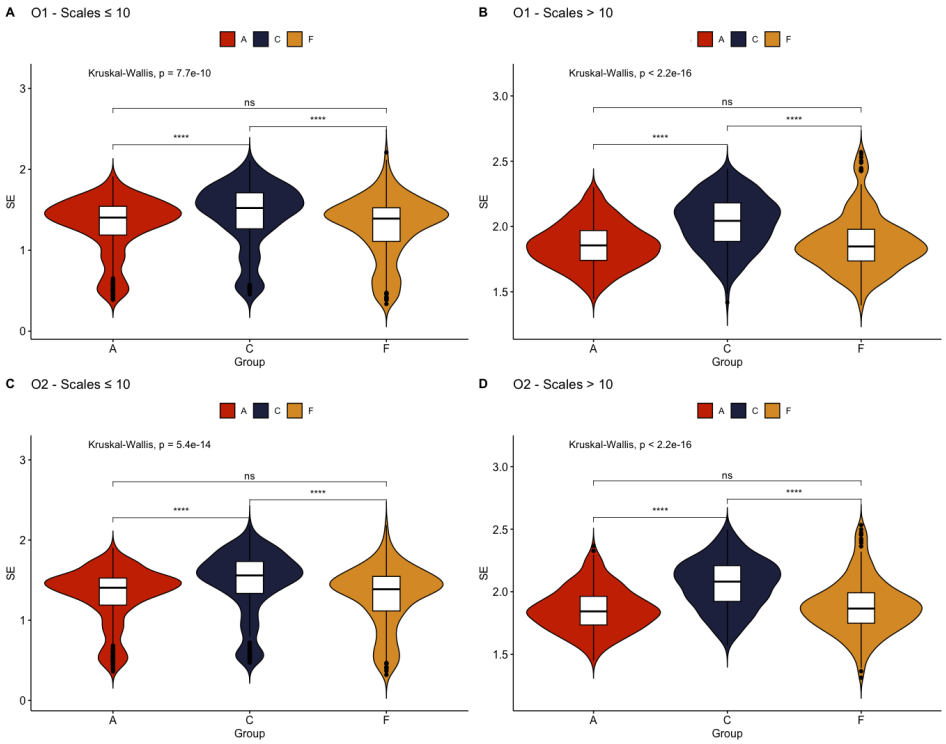


Figure 5: **Occipital electrodes MSE distributions.** The violins plots illustrate the MSE values in the O1 electrode (A,B) and O2 (C,D): the left plots include values from scale 1 to 10 (included), while the right plots show values from scale 11 to 20. The K-W values are shown together with the distribution of the values: it can be seen that no statistical difference is found between the two pathologies (Group A and F), but in all graphs there a statistical difference between healthy subjects (Group C) and AD/FtD patients, with healthy EEG data having higher entropy.